Reda El Ferrasi

Junior Data Scientist

+212 607420430 | redaelferrasi@gmail.com | linkedin.com/in/reda-elferrasi | github.com/redaelfer

PROFILE

State-qualified engineer in Data Science and IoT, with projects classifying security logs and medical images. Experience in data preparation and model evaluation, complemented by a Data Engineering internship at ONCF.

TECHNICAL SKILLS

Programming: Python, SQL.
Data: Pandas, NumPy, MongoDB, PostgreSQL.
ML: TensorFlow, feature engineering, model evaluation.
Tools: Docker, Git.
Languages: Arabic (native), English (B2), French (B2).

PROFESSIONAL EXPERIENCE

AI Engineer | Matious Digital – Final-year project (PFE) Jan. 2026 – Sep. 2026

- Developed and tested a platform with **20 to 40 concurrent AI agents**.
- Automated prompt generation, model selection, routing and error recovery.
- Integrated two agent runtimes through WebSocket and REST APIs.
- Deployed containerized agents on Hetzner Cloud.

Data Engineering Intern | ONCF Jun. 2025 – Aug. 2025

- Automated SNMP/Syslog ETL: reduced latency by **23 %** during testing.
- Centralized data and created dashboards with **at least 10 KPIs**.
- Reduced incident detection time by **35 %** during NOC testing.

Full Stack Developer Intern | S4U Jul. 2024 – Aug. 2024

- Developed a React ticketing SaaS: reduced processing time by **30 %** in tests.
- Added Jest tests with **72 % coverage** of key features.

PROJECTS

APT Attack Detection | AI and Cybersecurity | github.com/redaelfer/NosqlAPT

- Preprocessed DAPT 2020 (Kaggle): missing values, encoding and feature normalization.
- Simulation: F1 **0.70 to 0.73 (+0.3)** after adding autoencoder features to XGBoost.

RetinaScan | Medical AI | github.com/redaelfer/RetinaScan

- Prepared images from Retinal Disease Classification (Kaggle): normalization and augmentation.
- Simulation: accuracy **73 to 76% (+3 points)**, with image augmentation and dropout.

BigData | Streaming Data Pipeline | github.com/redaelfer/BigData

- Processed price data from a Python producer with Spark Structured Streaming.
- Stored data and checkpoints in HDFS using Docker.

EDUCATION

ENSIAS – State Engineering Degree – Data Science & IoT 2023 – 2026

CPGE – TSI preparatory classes 2021 – 2023

CERTIFICATIONS & INTERESTS

CS50x | Harvard University

AWS Knowledge: Cloud Essentials – Training badge | AWS

Interests: ECU remapping, car/motorcycle mechanics and repair, game modding, history/mythology.